

A 112 Gbps and Direct-drive Silicon Microdisk Modulator

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Abstract We present a silicon microdisk modulator with 30 GHz electro-optic bandwidth and 0.57 V·cm modulation efficiency. For the first time, the silicon microdisk modulator can be directly driven with 0.8 V_{pp} swing for 112 Gbps four-level pulse amplitude modulation. ©2026 The Author(s)

Introduction

The explosive growth of data throughput brought by the booming of artificial intelligence and machine learning has posed a serious challenge to interconnects in data center. Scale-up architecture requires significantly higher bandwidth density in single computing node [1]. Silicon photonics (SiPh) microring modulators (MRMs) and microdisk modulators (MDMs) are attractive candidates owing to their advantages of compact footprint, low energy per bit and compatibility to dense wavelength division multiplexing (DWDM) [2]. A DWDM optical transmitter can be realized by cascading multiple MRMs or MDMs carrying different wavelength channels via single bus waveguide, thus achieving high bandwidth density [3,4].

Benefiting from low radiative bend loss caused by the single cavity sidewall, MDMs can be designed with an ultra-small radius to achieve wider free spectral range (FSR) over 5 THz compared to MRMs [5,6]. A large FSR is essential for accommodating more wavelength channels, which is desirable to eventually improve the bandwidth density [7]. However, the smaller cavity size might cause quality factor (Q-factor) degradation and sacrifice the modulation efficiency. Several demonstrations of MDMs adopt vertical PN junction to improve the modulation efficiency, but the increased junction capacitance has become a limiting factor to the electro-optic (EO) bandwidth [8,9].

In this paper, we demonstrate a silicon MDM with a PN junction resembling L-shape for an optimal trade-off between modulation efficiency and EO bandwidth. The modulator achieves an FSR of 2.5 THz, a 3-dB EO bandwidth of 30 GHz and a high modulation efficiency of 0.57 V·cm. It supports up to 112 Gbps four-level pulse amplitude modulation (PAM-4) by only 0.8 V_{pp} direct drive without extra electrical amplifier.

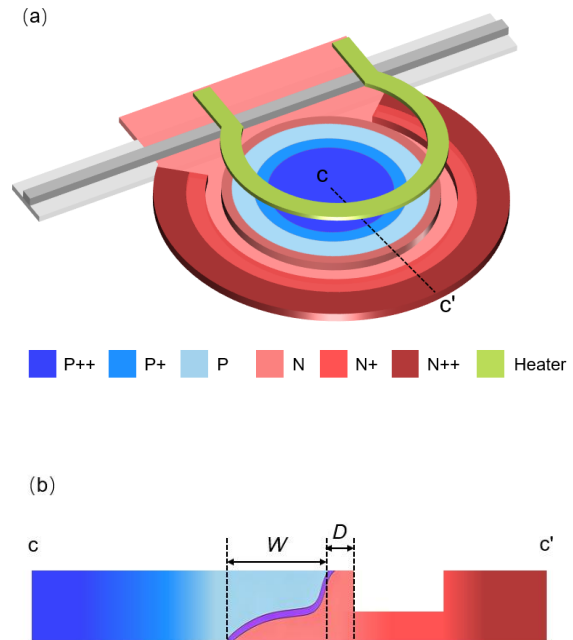


Fig. 1: (a) Schematic structure of the proposed MDM from the top view, (b) doping profile along the dash line from c to c'.

Device design

Fig. 1(a) shows the schematic structure of the proposed MDM from the top view. The device is designed on silicon-on-insulator (SOI) platform with 220 nm top silicon and 2 μm buried oxide. The radius of the MDM and the bus waveguide width are designed to be 5.3 μm and 450 nm respectively. To lower the sheet resistance, only partial silicon is etched to be 90 nm slab layer with 900 nm trench width along the cavity edge. Fig. 1(b) shows the doping profile along the dash line from c to c'. It can be seen that a curving PN junction resembling L-shape is formed inside the disk cavity using lightly doped P and N regions. This

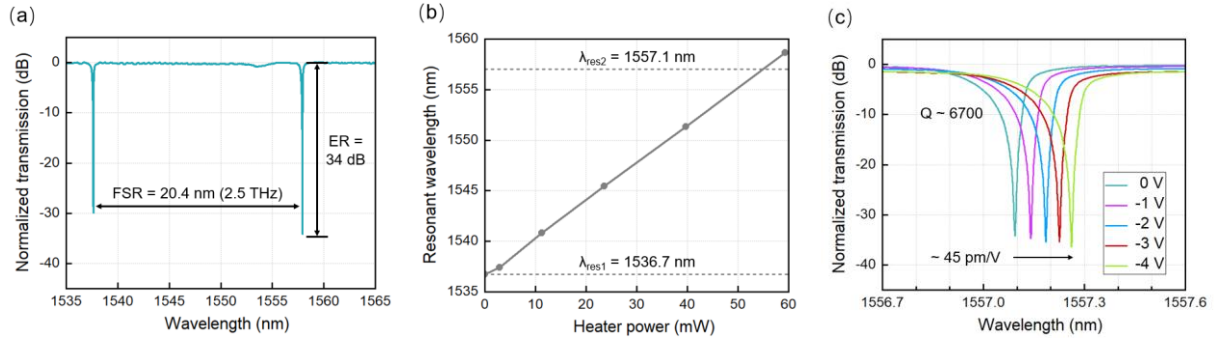


Fig. 2: (a) Normalized transmission spectrum of the fabricated MDM, (b) measured resonant wavelengths under various heater powers, (c) measured transmission spectra near the resonance of 1557.1 nm under different reversed bias.

PN junction is customized by adjusting the implant dose and energy used for both boron doping and phosphorous doping, which is totally enabled by the foundry process. Furthermore, the junction width W , and the distance between junction and cavity edge D are co-optimized to improve the overlap with the optical mode, as well as minimize the increase of junction capacitance, thus realizing an optimal trade-off between modulation efficiency and EO bandwidth. Optimized values of W and D for the PN junction are 650 nm and 120 nm respectively.

Device characterization

The designed MDM is fabricated on an 8-inch SiPh platform in a commercial foundry. Fig. 2(a) shows the normalized transmission spectrum of the fabricated MDM. The FSR is measured to be 20.4 nm (2.5 THz), and an extinction ratio (ER) over 30 dB indicates the MDM is very close to the critical coupling condition. A TiN heater is integrated on top of the cavity edge to thermally tune the resonant wavelength. The resonant wavelengths under various heater powers is shown in Fig. 2(b). The measured heater efficiency is about 0.37 nm/mW, which also implies that 5 V is able to thermally tune the resonance across a full FSR. Fig. 2(c) shows the measured transmission spectra near the resonance of 1557.1 nm under different reversed bias. A loaded Q-factor is estimated to be 6700. An average electrical tuning efficiency of 45 pm/V is measured, which corresponds to a modulation efficiency of 0.57 V·cm.

The dynamic EO responses of the MDM was then characterized by a 67 GHz vector network analyzer (VNA). The RF signal and reversed bias were applied to the MDM via a 67 GHz ground-signal-ground (GSG) probe. A calibration substrate was used to factor out the responses from the probe and cables. Fig. 3(a) shows the measured EO responses under various reversed bias with 5 dB insertion loss (IL) detuning level. The results indicate a 3-dB bandwidth of 30 GHz is

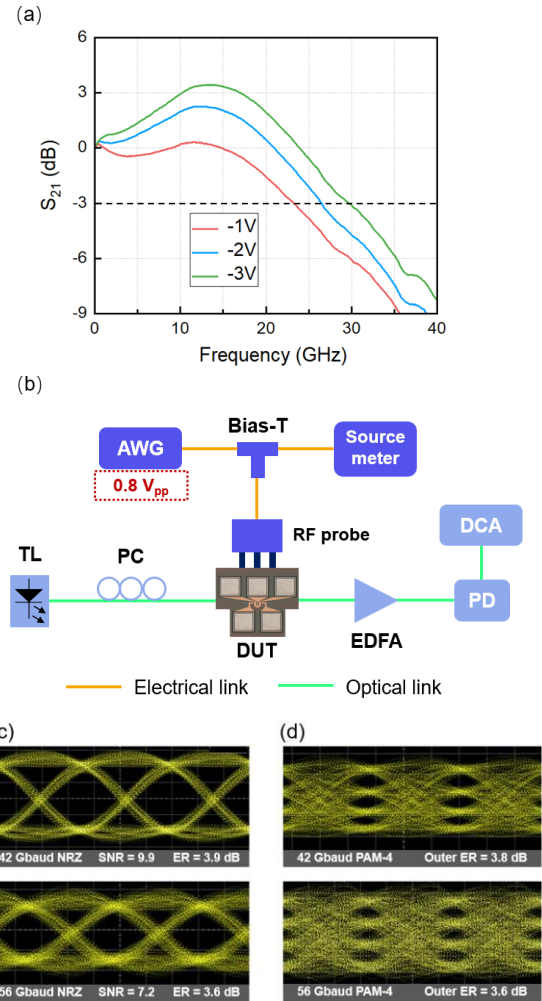


Fig. 3: (a) Measured EO responses under various reversed bias with 5 dB IL detuning level, (b) Experimental setup of high-speed characterization of the fabricated MDM, (c) measured eye diagrams of 42 and 56 Gbaud NRZ, (d) measured eye diagrams of 42 and 56 Gbaud PAM-4.

achieved under -3 V reversed bias, which represents a benchmark record of the silicon MDM. Fig. 3(b) shows the experimental setup for characterizing the high-speed modulation performance of the fabricated MDM. A tunable laser (TL) was

Tab. 1: Comparison with other state-of-the-art MDMs

Ref	Band	Junction type	FSR (THz)	Efficiency (pm/V)	BW (GHz)	Voltage swing (V_{pp})	Data rate (Gbps)
[5]	O	Vertical	5.4	55	19.5	0.8	40 (NRZ)
[6]	O	Vertical	6.4	68	25.8	1.6	50 (NRZ)
[8]	C	Vertical	3.4	58	16	0.8	32 (NRZ)
[9]	C	Vertical	7.3	43	—	1.3	16 (NRZ)
[10]	C	Lateral	3.5	8	30.5	7.6	25 (NRZ)
This work	C	Vertical	2.5	45	30	0.8	112 (PAM-4)

used to adjust the operating wavelength instead of thermal tuning for proof-of-concept demonstration. The laser output was aligned to TE mode through a polarization controller (PC) and coupled to the MDM chip. The RF signal was generated by a 224 GSa/s arbitrary waveform generator (AWG) and directly connected to the bias tee. The swing output is set to be $0.8 V_{pp}$ to directly drive the modulator without using an extra electrical amplifier. A reversed bias from source meter was also sent to the bias tee. The output power of the modulated optical signal was first amplified by an Erbium-Doped Fiber Amplifier (EDFA) to compensate for the link loss. Then the modulated signal was received by a high-speed photodetector (PD) and finally captured by a digital communication analyzer (DCA). Fig. 3(c) and 3(d) show the measured eye diagrams of NRZ and PAM-4 modulation with different baud rates respectively. A reversed bias of -3 V and detuning level of 5 dB IL were used for both NRZ and PAM-4 modulation. Clear and opened eye diagrams of 42 Gbaud and 56 Gbaud NRZ signals were obtained with SNR over 7 and ER over 3.6 dB, while eye diagrams of corresponding PAM-4 modulation were obtained with outer ER over 3.6 dB.

Table 1 summarizes the recent progress of state-of-the-art demonstrations on silicon MDMs in O-band and C-band. We have achieved a moderate modulation efficiency among these works, while the bandwidth is larger than most of them. Though ref. [10] also achieved high bandwidth similar to us, the driving voltage in this work is significantly smaller. We have achieved a record high bandwidth with direct-drive level in the meantime. Owing to an optimal design of the PN junction, the MDM is capable of realizing high-speed direct-drive modulation beyond 100 Gb/s for the first time.

Conclusion

We presented and experimentally demonstrated a silicon MDM with 30 GHz EO bandwidth and $0.57 V \cdot cm$ modulation efficiency. The PN junction is designed to realize an optimal trade-off between modulation efficiency and EO bandwidth.

It supports 112 Gb/s PAM-4 modulation via direct drive of only $0.8 V_{pp}$. The device shows a great potential for energy-efficient and high-speed inter-chip interconnections.

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